1. INTRODUCTION TO RESISTORS

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Resistors are the most commonly used component in electronics and their purpose is to create specified values of current and voltage in a circuit. A number of different resistors are shown in the photos. (The resistors are on millimeter paper, with 1cm spacing to give some idea of the dimensions). Photo 1.1a shows some low-power resistors, while photo 1.1b shows some higher-power resistors. Resistors with power dissipation below 5 watt (most commonly used types) are cylindrical in shape, with a wire protruding from each end for connecting to a circuit (photo 1.1-a). Resistors with power dissipation above 5 watt are shown below (photo 1.1-b).

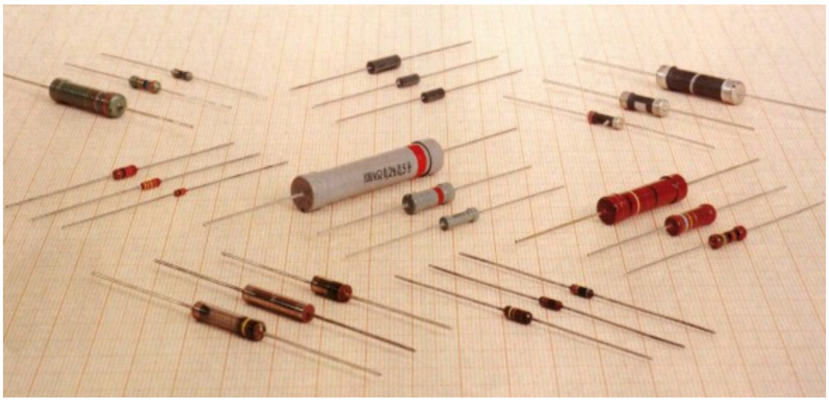

Fig. 1.1a: Some low-power resistors

Fig. 1.1b: High-power resistors and rheostats

The symbol for a resistor is shown in the following diagram (upper: American symbol, lower: European symbol.)

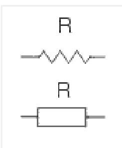

Fig. 1.2a: Resistor symbols

The unit for measuring resistance is the **OHM**. (the Greek letter Ω – called Omega). Higher resistance values are represented by “k” (kilo-ohms) and M (meg ohms). For example, 120 000 Ω is represented as 120k, while 1 200 000 Ω is represented as 1M2. The dot is generally omitted as it can easily be lost in the printing process. In some circuit diagrams, a value such as 8 or 120 represents a resistance in ohms. Another common practice is to use the letter E for resistance in ohms. The letter R can also be used. For example, 120E (120R) stands for 120 Ω, 1E2 stands for 1R2 etc.